

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (Biotechnology) (Credit System) Degree Examinations Supplementary Examinations - December 2025

21BT405 / 20BT305 - BIOPROCESS PRINCIPLES AND CALCULATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define the following: i) Normality ii) Molarity iii) Molality iv) PPM	8	L*2	1	1,2, 6,7
	b) A mixture has 30 moles of A, 25 moles of B and 60 moles of C. Molecular weights of A, B & C are 28, 44 and 60 respectively. Calculate mole% and mass % of each component.	8	L2	1	1,2, 6,7
	c) What is the Carnot efficiency of an engine operating between thermal reservoirs of temperature 400 K and 280 K? Calculate the maximum work done by this engine for heat supply of 100J.	4	L2	1	1,6
2.	a) An aqueous solution of H ₂ SO ₄ is prepared by dissolving 43 kg of H ₂ SO ₄ in 100kg of water at 293K. Find the normality and molarity.	6	L2	1	1,2, 6,7
	b) Determine the pressure exerted by O ₂ in a container of 2 m ³ capacity when it contains 160 moles at 27°C using: i) Ideal gas equation ii) Vander Waal's equation Data: $a = 139.25 \frac{KNm^4}{Kmol^2}$ $b = 0.0314 m^3/kmole$	10	L2	2	2
	c) What is thermodynamic system? What are the different types?	4	L2	2	1,2
3.	a) A reversible heat engine takes heat at the rate of 500kJ/s from a heat source at 700K. The work done by the cyclic device is 200 kJ/s and rejects heat to sinks at 400K. Calculate i) the engine thermal efficiency ii) amount of heat rejected.	10	L2	2	2,3
	b) Define Kelvin-Planck and Clausius statements of second law of thermodynamics.	4	L2	2	2
	c) Distinguish between the following: i) State function and path function ii) Intensive and extensive properties iii) Mechanical and thermal equilibrium.	6	L1	2	1
Unit – II					
4.	a) One hundred kilograms of dilute waste acid containing 30% sulfuric acid is to be concentrated. 50% sulfuric acid using concentrated sulfuric acid of strength 96%. How many kilograms of concentrated sulfuric acid is required for this purpose?	8	L4	3	1,2
	b) A tank of weak battery acid contains 12.43% H ₂ SO ₄ . If 200kg of 77.7% H ₂ SO ₄ is added to the tank and the final solution contains 18.63% H ₂ SO ₄ , how many kgs of final battery acid have been made?	8	L2	3	2,3

P.T.O.

- c) Draw a schematic block diagram for liquid liquid extraction operation. Write material balance equations. Assume that solvent is pure. Feed contains one solvent dissolved in feed solvent.
- 4 L3 3 1,2
5. a) A feed containing 50% Benzene and 50% Toluene is fed to a distillation column at the rate of 5000kg/hr. The top product containing 95% Benzene and the bottom product containing 8% Benzene on a weight basis. Calculate i) the mass flow rate of distillate and residue ii) % recovery of Benzene.
- 10 L2 3 2,3
- b) A dryer system handles 2500kg of wet solid/day containing 50% solid and 50% moisture, are fed to first dryer. The product comes out has 20% moisture. This is admitted to the 2nd dryer from where the product coming out has 2% moisture. Calculate % of original water removed in each dryer and the final weight of product dryer.
- 10 L3 3 2,3
6. a) Briefly explain general energy balance equation for fermenter and heat exchangers.
- 10 L2 4 2,3, 6,7
- b) In the range of 200 K to 600 K, NH_3 has a heat capacity, $C_p = a + bT + cT^2 + dT^3 \frac{\text{KJ}}{\text{Kmol.K}}$, where
 $a = 20.1494$,
 $b = 845.765 \times 10^{-3}$
 $c = -4067.45 \times 10^{-6}$
 $d = 6606 \times 10^{-9}$.
 Calculate the mean heat capacity of NH_3 in the given temperature range.
- 10 L3 4 2,3
- Unit – III**
7. a) Briefly explain the following with example.
 i) Limiting reactant ii) Conversion iii) Yield iv) Selectivity
 v) % excess
- 10 L2 5 2,3, 6
- b) In the production of chlorine gas by the oxidation of HCl gas, air is used 30% in excess of that theoretically required. Based on the 4 K moles of HCl, and reaction is 80% complete, calculate the composition of product.
- 10 L4 5 2,3
8. a) Write a note on the following.
 i) Standard heat reaction ii) Heat of formation
 iii) Heat of combustion
- 6 L2 5 2,3
- b) Find the degree of reduction for the following:
 i) Glucose ii) Ethanol($\text{C}_2\text{H}_5\text{OH}$) iii) Carbon dioxide
 iv) Methanol (CH_3OH)
- 4 L2 5 2,3
- c) Calculate ΔH°_R for the following reaction
 $\text{C}_2\text{H}_5\text{OH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$
 Data: ΔH°_c (cal)
 $\text{C}_2\text{H}_5\text{OH} = -326700$ $\text{CH}_3\text{COOH} = -208340$
 $\text{CH}_3\text{COOC}_2\text{H}_5 = -538760$ $\text{H}_2\text{O} = 0$.
- 10 L2 5 2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
